

# IBM University Engagement Project Submission

*“Exploring the Power of Agentic AI with IBM Granite and LangFlow”*

## Project Tile – [Smart Study Generator Agent]

### Domain of Project – [ Education ]

| Student Name      | Recent Passport Photo                                                              | Email ID                | Mobile number [WhatsApp] |
|-------------------|------------------------------------------------------------------------------------|-------------------------|--------------------------|
| Zikra Azam Pathan |  | zikrapathan07@gmail.com | 7620125144               |

# Problem Statement -

## The Challenge

Students today face an overwhelming flood of information—lecture notes, textbooks, research papers, YouTube tutorials, and online resources. This content is scattered, unorganized, and often not personalized to their learning pace or style. As a result, students struggle with finding the right study material, summarizing key concepts, and staying exam-ready. Traditional study tools only provide static notes or flashcards, failing to adapt dynamically to student needs. The challenge is to move beyond static resources toward an intelligent, adaptive study companion.

## The Objective

Build an agentic AI-powered Smart Study Generator that personalizes and streamlines learning for students. The solution should:

- 1. Fuse Learning Resources:** Aggregate content from class notes, textbooks, research articles, and online tutorials into structured, easy-to-digest study material.
- 2. Enable Multimodal Input:** Allow students to upload notes, voice queries, or images (e.g., textbook pages), and generate concise summaries, flashcards, or concept maps.
- 3. Create a Predictive & Agentic Layer:** Recommend personalized study plans, highlight important topics based on syllabus or past exams, and provide predictive alerts (e.g., “Focus on Topic X—frequently asked in exams”).
- 4. Visualize Study Progress:** Offer a real-time dashboard showing learning milestones, weak areas, practice quizzes, and performance forecasts.

# Proposed Solution -

The proposed solution is to develop an AI-powered Smart Study Generator Agent using Langflow and the Gemini Language Model. The system acts as a virtual study assistant that helps students understand concepts, improve learning efficiency, and prepare effectively for exams. It provides instant, accurate, and personalized academic support through an interactive chatbot interface.

## Key Features of the Proposed Solution:

-  Provides clear and simple explanations for academic topics.
-  Generates concise study notes and topic-wise summaries.
-  Creates personalized study plans based on students' learning goals.
-  Generates quizzes, MCQs, and practice questions for self-assessment.
-  Helps students with quick revision by highlighting important concepts.
-  Answers students' questions in natural language with accurate responses.
-  Improves learning experience by making study sessions more interactive and engaging.
-  Reduces students' dependency on searching multiple online resources by providing instant academic assistance in one place.

The entire workflow is designed visually using Langflow, while the Gemini Language Model powers the intelligent responses. This solution is user-friendly, scalable, and can be enhanced in the future with features such as PDF-based learning, multilingual support, voice interaction, and student progress tracking.

## Technology Used

- **1. Langflow** : Used to design and build the AI agent workflow using a visual interface.
- **2. Google Gemini 3.5 Flash** : Used as the Large Language Model (LLM) to understand user queries and generate intelligent responses.
- **3. Google AI Studio API** : Used to connect the Gemini model with the Langflow application.

## Langflow component Used -

- 1.Chat Input** : Receives questions and study-related requests from the student.
- 2. Agent** : Acts as the core AI assistant that processes user queries and generates responses.
- 3. Google Gemini 3.5 Flash (Language Model)** : Powers the agent by understanding queries and generating intelligent responses.
- 4. Agent Instructions** : Defines the behavior, responsibilities, and response style of the Smart Study Generator Agent.
- 5. Chat Output** : Displays the generated responses to the user.

# Langflow workflow -

The screenshot displays the Langflow web interface in a browser window. The address bar shows the URL `localhost:7860/flow/ae96c4ab-e2f4-4e74-863a-6133411c0ed6`. The interface includes a sidebar on the left with a search bar containing the text "language". Below the search bar, there are sections for "Components" (Models & Agents, Language Model, Policies Beta) and "Bundles" (IBM, IBM watsonx.ai, xAI, vLLM, Vertex AI, SambaNova). The main workspace features a workflow diagram on a dark grid background. It starts with a "Chat In..." node (28ms) connected to an "Agent" node. The "Agent" node is configured with the "Language Model" set to "gemini-3.5-flash", "Agent Instructions" set to "You are Smart Study Generator Age...", and "Tools" set to "Receiving input". The "Agent" node's "Response" output is connected to a "Chat Output" node. The top right of the interface shows "Starter Project / Smart Study Generator Agent" and social media links for 150k GitHub followers and 25k Discord members. A "Playground" button and a "Share" dropdown are also visible.

language

Components

- Models & Agents
- Language Model
- Policies Beta

Bundles

- IBM
- IBM watsonx.ai
- xAI
- xAI
- vLLM
- vLLM
- Vertex AI
- Vertex AI
- Vertex AI Embeddin...
- SambaNova

New Custom Component

Chat In... 28ms

Agent

Define the agent's instructions, then enter a task to complete using tools.

Language Model\* ⓘ

gemini-3.5-flash

Agent Instructions ⓘ

You are Smart Study Generator Age...

Tools ⓘ

Input ⓘ

Receiving input

Response

Chat Output

150k 25k

Playground Share

99%

# Out put of the agent:

Default Session



User

give me 5 days of study plan



AI

✓ Finished in

🗣️ 1.5K | 12.4s

I would love to help you create a personalized study plan!

To make this plan as effective as possible, **could you tell me:**

1. **What subject, topic, or exam** are you preparing for?
2. **How many hours** can you dedicate to studying each day?

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In the meantime, here is a highly effective, versatile **5-Day Study Plan Template** (using active recall and spaced repetition) that you can apply to almost any subject:



## The Ultimate 5-Day Study Framework

Z



Default Session 

User

please explain me the concept of python dictionary



AI

✓ Finished in

 2.5K | 12.0s

A **Python Dictionary** is one of the most useful and powerful data structures in Python.

To understand it easily, let's use a real-world analogy!

### The Real-World Analogy

Think of a **real-life language dictionary**:

- You look up a **word** (e.g., "Apple").
- You find its **definition** (e.g., "A round red or green fruit").

In Python, a dictionary works exactly the same way!



- You find its **definition** (e.g., "A round red or green fruit").

In Python, a dictionary works exactly the same way!

- The **Word** is called a **Key**.
- The **Definition** is called a **Value**.

Together, they form a **Key-Value Pair**.

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## How it Looks in Python

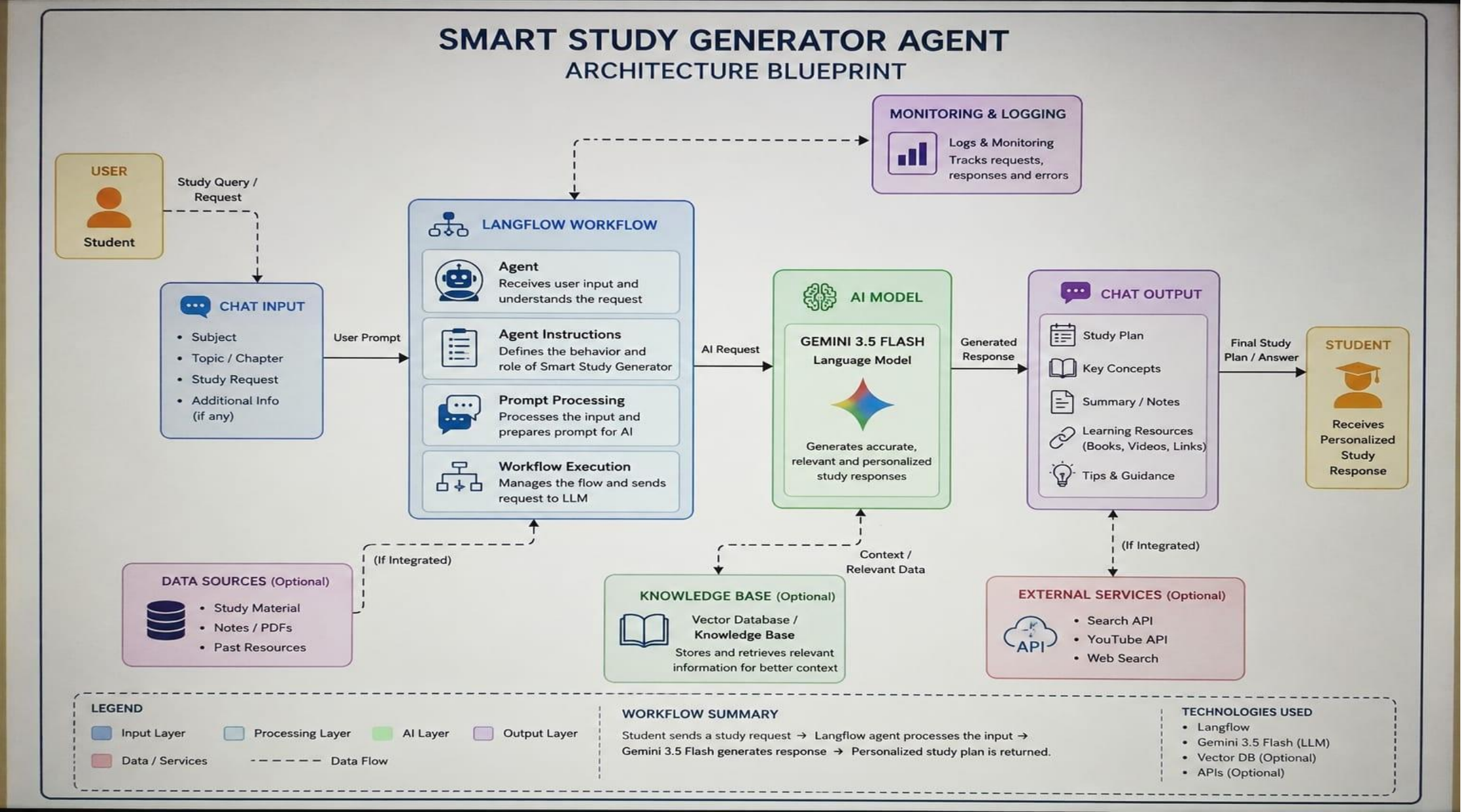
In Python, dictionaries are written inside **curly braces**

```
{ }
```

, and keys are separated from their values by a **colon**



# Architecture blueprint



# Role of Agentic AI in the solution

The Smart Study Generator Agent uses Agentic AI to act as an intelligent virtual study assistant. Instead of only responding to questions, the agent understands the student's learning needs and provides personalized academic support. It helps students study more efficiently by generating useful learning resources and guiding them throughout their learning journey.

## Role of Agentic AI

- ❖ Understands students' academic questions and learning requirements.
- ❖ Generates intelligent, context-aware, and accurate responses using the Gemini language model.
- ❖ Creates concise study notes and topic summaries.
- ❖ Generates quizzes, MCQs, and practice questions for self-assessment.
- ❖ Suggests personalized study plans based on students' goals.
- ❖ Explains complex concepts in simple and easy-to-understand language.
- ❖ Encourages interactive and self-paced learning through conversational AI.
- ❖ Provides instant academic assistance, reducing the time required to search for study materials.

Overall, Agentic AI makes the learning process more personalized, interactive, and efficient by acting as a smart educational assistant that supports students throughout their studies.



## Novelty and Uniqueness

- ✓ Agentic AI-Based Learning Assistant for intelligent and interactive student support.
- ✓ Hyper-Personalized Learning by providing customized study plans, notes, and explanations.
- ✓ Real-Time Feedback with instant responses to students' academic queries.
- ✓ AI-Powered Content Generation for summaries, quizzes, MCQs, and revision notes.
- ✓ Natural Language Interaction for a simple and user-friendly learning experience.
- ✓ Future-Ready Architecture with support for RAG-based document retrieval, Multi-Agent Intelligence, and Multimodal Inputs (PDF, Voice, Images).
- ✓ Scalable and Easy to Enhance using Langflow's visual workflow, allowing new AI features to be integrated in the future.

## Git Hub Link

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# Future Scope

**PDF-Based Learning Support** – Enable the agent to analyze uploaded study materials and answer questions directly from documents using RAG.

**Voice-Based Interaction** – Allow students to ask questions through voice commands and receive spoken responses.

**Multilingual Support** – Support multiple languages such as English, Hindi, Marathi, and Urdu for better accessibility.

**Student Progress Tracking** – Monitor learning progress, quiz performance, and provide personalized improvement suggestions.

**Adaptive Learning** – Recommend study plans and learning materials based on the student's performance and learning pace.

**Multi-Agent Collaboration** – Introduce specialized AI agents for note generation, quiz creation, doubt solving, and study planning.

**Multimodal Learning** – Support text, PDFs, images, and handwritten notes for a richer learning experience.

**Cloud Deployment** – Deploy the application on cloud platforms to make it accessible anytime and from any device.

# Certificates Earned:

In recognition of the commitment to achieve professional excellence



Zikra Pathan

Has successfully satisfied the requirements for:

Getting Started with Artificial Intelligence



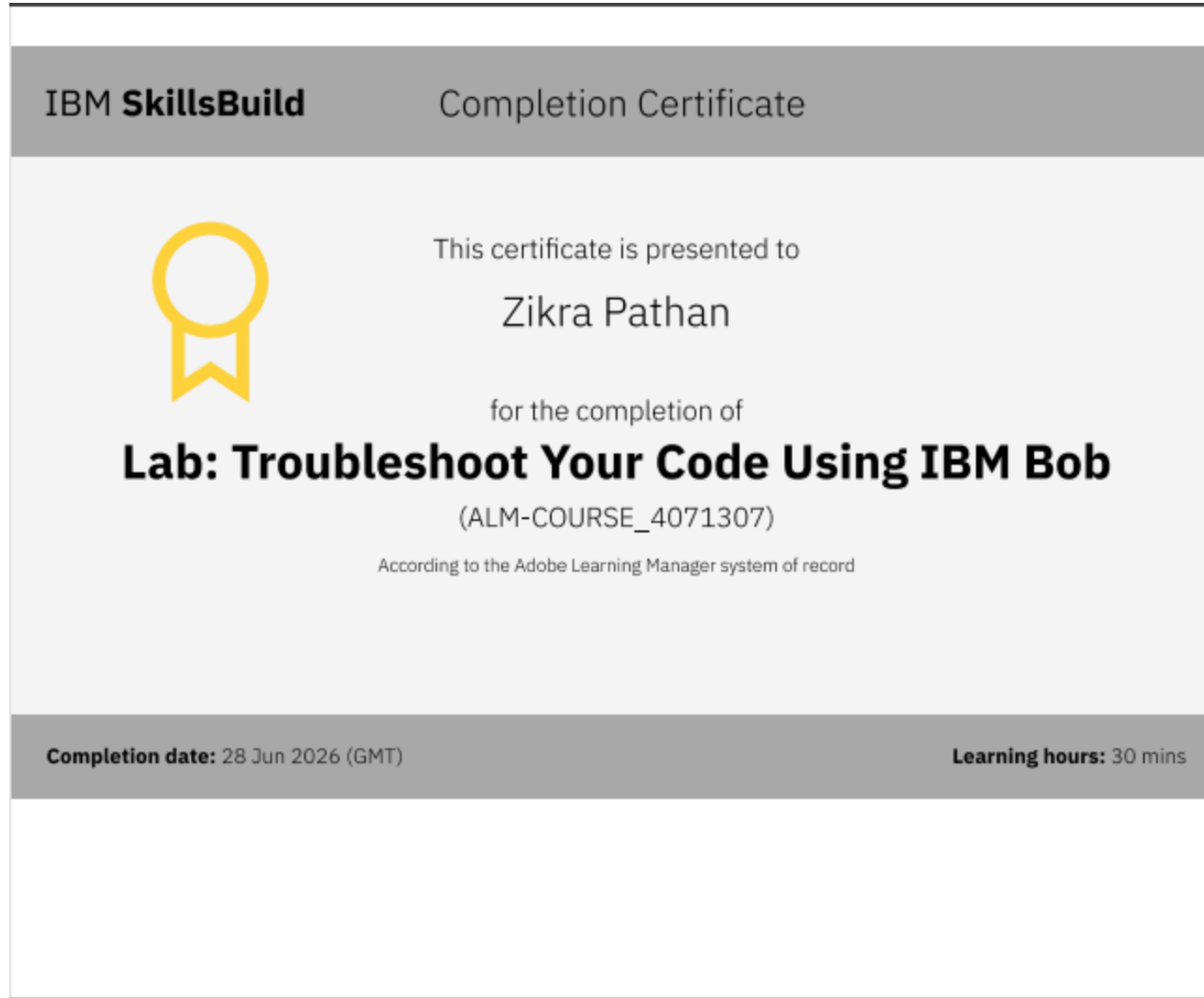
Issued on: Jun 25, 2026  
Issued by: IBM SkillsBuild

Verify: <https://www.credly.com/badges/0bae0516-38d2-447d-b5cb-3b8fcab03eb5>



# Certificates Earned:

## Add your IBM BOB Certificate



# Thank You!

Thank you for your time and interest.